

Faculty of Science & Technology
PES's Modern College of Engineering
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Level 7: First Year M. Tech. (2024 Pattern)

Course Code	CSE08651
Course Name	Skill Development – II
Semester	IV
Experiment No.	5
Title	Research Workflow Automation with GitHub

Problem Statement

To study and analyze the use of version control systems for automating and managing research workflows and collaborative development.

Objective

To make use of GitHub for automating research workflows, version control, and collaborative management of research artifacts.

Outcome

CO653.5: Apply version control tools to automate and manage collaborative research workflows.

Software / Hardware Requirements

- GitHub Platform
- Git Version Control System
- Web Browser
- Internet Connectivity
- Personal Computer or Laptop

Theory

Research workflow automation involves the systematic management of research activities such as data handling, code development, documentation, and collaboration. As research projects grow in complexity, maintaining consistency and traceability of changes becomes challenging. Version control systems address these challenges by tracking modifications and supporting collaborative development.

Git is a distributed version control system that enables researchers to track changes in files, manage multiple versions of documents, and collaborate effectively. It allows users to maintain a complete history of modifications, revert to previous versions, and merge contributions from multiple collaborators. These features are particularly useful in research environments where reproducibility and transparency are critical.

GitHub is a web-based platform built on Git that provides additional tools for repository management, collaboration, and workflow automation. It supports features such as issue tracking, pull requests, and continuous integration. By using GitHub, researchers can automate research workflows, manage code and documentation efficiently, and ensure organized and reproducible research practices.

Steps Involved

1. A GitHub account was created and configured.
2. A new repository was created for the research project.
3. Research files and documentation were added to the repository.
4. Version control operations such as commit and push were performed.
5. Collaboration features such as issues and pull requests were explored.

Results

The research workflow was successfully managed using GitHub. Version control and automation features enabled efficient tracking of changes and collaborative research management.

Analysis and Conclusion

Analysis: GitHub provided an effective platform for automating research workflows by enabling version control, collaboration, and change tracking across research artifacts.

Conclusion: The use of GitHub for research workflow automation enhances collaboration, ensures reproducibility, and supports systematic management of research projects.

References

1. GitHub, “GitHub Documentation,” Available: <https://docs.github.com>
2. Chacon, S., and Straub, B., *Pro Git*, Apress.